

Member 2 – Backend Integration Contract

Purpose

This document defines the exact backend requirements and integration contracts between Member 2 (Backend) and Members 1, 3, and 4 (Tracking App, ML Engine, Dashboard). Following this contract ensures zero refactor and seamless integration.

1. What We Will Give Member 2 (To Make Backend Build Easy)

- Frozen list of event types (no future renaming).
- Role-to-event permission matrix for validation.
- Clear separation of responsibilities (Backend ≠ ML ≠ Dashboard).
- Database expectations: events are raw & immutable; digital_twins are derived.
- Guaranteed event payload structure from frontend.
- ML processing assumptions (polling, processed flag usage).
- Security expectations (JWT, role enforcement, ownership checks).

Frozen Event Types

- attendance_marked
- test_score_recorded
- activity_completed
- teacher_attendance
- marks_uploaded
- behavior_observation
- parent_feedback
- acknowledged_alert
- system_configuration_change

Role → Event Permissions

- Student → attendance_marked, test_score_recorded, activity_completed
- Teacher → teacher_attendance, marks_uploaded, behavior_observation
- Parent → parent_feedback, acknowledged_alert
- Admin → system_configuration_change

2. What Member 2 Must Give Us (For Proper Integration)

- Base API URL (dev / prod).
- Authentication endpoints (login, signup) returning JWT.

- JWT expiry time and role embedding.
- Profile mapping endpoint (/profile/me) with role-specific response shapes.
- Event ingestion endpoint (/events) accepting raw events only.
- Read-only digital twin endpoints for dashboard.
- Consistent error response format and HTTP status codes.
- CORS configuration for frontend domains.

Mandatory Endpoints

- POST /auth/login
- POST /auth/signup
- GET /profile/me
- POST /events
- GET /twins/self
- GET /twins/child/{child_id}
- GET /twins/class/{class}/{section}
- GET /twins/all

Error Handling Contract

Backend must return standard HTTP status codes: 401 (Unauthorized / Token Expired), 403 (Forbidden / Role Violation), 400 (Validation Error), 500 (Server Error). Error response body should contain a readable 'detail' field.

Final Note: This backend is a contract provider, not a business logic engine. If this document is followed, Members 1, 3, and 4 will integrate without any redesign.